

Effects of grass type on hydraulic response of the three-layer landfill cover system

Song Feng^{1,2} , Hong Wei Liu^{3,4} , Qi Peng Cai⁵
and Wen Bin Jian⁶

Waste Management & Research
2022, Vol. 40(7) 882–891
© The Author(s) 2021
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/0734242X211061213
journals.sagepub.com/home/wmr


Abstract

Soil column tests were conducted to investigate the effects of grass type on water infiltration in a three-layer landfill cover under drying and wetting conditions. Five soil columns were prepared, including one bare, two Bermuda grass-planted and the other two vetiver-planted. During the drying period, the suction of vetiver-planted soil column was the largest, while that of bare case was the lowest. During the wetting period, the infiltration rate shows a bimodal form due to the contrasting hydraulic properties of different soil layers. The infiltration rate of vetiver-planted soil column was the lowest, followed by Bermuda grass-planted and bare cases. Correspondingly, the vetiver-planted soil column retained the maximum suction and the deepest ponding depth during rainfall. This was likely due to the larger leaf area and deeper roots of vetiver than those of Bermuda grass, thus inducing the maximum initial suction by root water uptake before rainfall and reducing the water permeability by root occupations of soil pores. These results show that vetiver is more effective than Bermuda grass to reduce water percolation through the three-layer landfill cover.

Keywords

Grass type, soil column, suction, water infiltration, three-layer landfill cover

Received 21st May 2021, accepted 2nd November 2021 by Associate Editor David E. Ross.

Introduction

Vegetation affects the hydraulic response of the geotechnical infrastructure, such as slope, levee and landfill cover (Białowiec et al., 2007; Khalilzad et al., 2014; Liang et al., 2020; Preston and McBride, 2004; Smethurst et al., 2006, 2015). The hydraulic effects of vegetation include (1) soil suction increases due to root water uptake, leading to a reduction in soil water permeability and hence less rainfall infiltration (Feng et al., 2019, 2020; Liu et al., 2016; Ng et al., 2015; Ng and Leung, 2012; Ng and Menzies, 2007; Woon, 2013) and (2) change of soil water permeability caused by the occupation of soil pores by roots (Fatahi et al., 2009; Garg et al., 2015; Ng and Zhan, 2007).

Extensive studies have been conducted to investigate the effects of vegetation on the hydraulic response of soil (Camerini and Groppali, 2014; Liu et al., 2018; Ni et al., 2018; Singh et al., 2018). Experimental studies show that plants could either reduce or increase water infiltration, depending on the soil type and plant characteristics (Fatahi et al., 2009; Gadi et al., 2018; Garg et al., 2015; Ng et al., 2016b, 2016c; Ng and Zhan, 2007; Woon, 2013). However, these studies mainly focus on the effects of vegetation in a single layer soil. Multi-layer soils are frequently encountered in engineering practice, such as the traditional landfill cover with capillary barrier effects (CCBE), which consists of a fine-grained soil layer overlying a coarse-grained one. Recently, Ng et al. (2016a) proposed a three-layer landfill cover by incorporating a compacted clay layer underneath a CCBE. Both

laboratory (Ng et al., 2016a) and field tests (Ng et al., 2019a) demonstrated that this three-layer landfill cover could work satisfactorily in a humid region (i.e. annual rainfall precipitation >2000 mm) in terms of preventing rainfall percolation. Grasses are commonly planted in landfill cover for ecological restoration. Yet, very few experimental studies are available for investigating the effects of vegetation on the hydraulic response in multi-layer soils. Ng et al. (2019b) found that shrub (*Schefflera arboricola*) was more effective in reducing water percolation than grass (*Cynodon dactylon*) through ponding test, in which a constant ponding depth was applied at the soil surface. Thus, no information about the temporal variation of ponding depth during rainfall as occurred in reality could be obtained. Moreover, the plant

¹MOE Key Laboratory of Soft Soils and Geoenvironmental Engineering, Department of Civil Engineering, Zhejiang University, Hangzhou, China

²College of Civil Engineering, Fuzhou University, Fuzhou, China

³Zijin School of Geology and Mining, Fuzhou University, Fuzhou, China

⁴Key Laboratory of Geohazard Prevention of Hilly Mountains, Ministry of Natural Resources (Fujian Key Laboratory of Geohazard Prevention), Fuzhou, China

⁵College of Civil Engineering, Huaqiao University, Xiamen, China

⁶Zijin School of Geology and Mining, Fuzhou University, Fuzhou, China

Corresponding author:

Hong Wei Liu, Zijin School of Geology and Mining, Sichuan Highway Planning, Survey, Design and Research Institute Ltd. and Southwest Jiaotong University Fuzhou University, Fuzhou 350000, Fujian, China. Email: hliuan@connect.ust.hk

Table 1. Basic properties of the soils.

Property	CDG	Silt	Gravelly sand	Clay
Unified soil classification system ^a	SM	ML	SP	CH
Specific gravity, G_s ^b	2.60	2.61	2.62	2.52
Atterberg limits ^c				
Liquid limit (LL)	33	22	–	59
Plastic limit (PL)	23	16	–	32
Plastic index (PI)	10	6	–	27
Standard compaction curve ^d				
Maximum dry density, ρ_d (kg m ⁻³)	1860	1710	1494	1260
Optimum water content (%)	15.5	18.0	–	36.2
Void ratio, e	0.60	0.69	0.58	1.06
Saturated water permeability, k_s (m s ⁻¹) ^{e,f}	5.0×10^{-6}	1.4×10^{-6}	9.7×10^{-3}	9.0×10^{-9}
Volumetric water content at saturation, w (%)	23.1	26.4	22.1	42.1

CDG: completely decomposed granite; SM: silty sand; MLL represents silt; SP: poorly graded sand; CH: Fat clay..

^aUsing the ASTM D2487 standard test method (ASTM, 2011) to unified Soil Classification System (USCS).

^bSpecific gravity tests were performed in accordance with the ASTM D854 standard test method (ASTM, 2010a).

^cAtterberg limit tests were performed using the standard test method ASTM D4318 (ASTM, 2010b).

^dCompaction curves for the CDG, silt and clay were determined in accordance with ASTM D698 (ASTM, 2012).

^eFlexible wall permeameter, as described in ASTM D5084 (ASTM, 2010c), was used for determining the saturated permeability (k_s) of the CDG, silt and clay.

^fConstant-head method, as described in ASTM D2434 (ASTM, 2006), was used for the gravelly sand.

characteristic was not quantified by Ng et al. (2019b) and hence its effects on the hydraulic response of multi-layer soil are not clear. Currently, there is a gap in the understanding of the influence of grass types and their characteristics on rainfall infiltration and pore water pressure distribution in multi-layer landfill cover.

This study aimed to experimentally investigate the effects of grass type on the hydraulic response of the three-layer landfill cover. In total, five soil columns were prepared, including one bare, two planted with Bermuda grass and the other two planted with vetiver. After adaptation period for plant growth, drying and wetting tests were conducted successively. During the tests, the measurements include suction, volumetric water content, water infiltration, ponding depth and plant characteristics.

Material and method

Testing soil and plant

Four types of soils were used in the present study, including clay, gravelly sand, silt and completely decomposed granite (CDG, silty sand). The properties of these soils are summarised in Table 1. Bermuda grass and vetiver were selected, both of which are commonly found in many parts of Asia (Hau and Corlett, 2003), including Hong Kong, Malaysia and Thailand. Bermuda grass and vetiver are widely cultivated for ecological restoration and rehabilitation (Beyrouthy et al., 1990; Hau and Corlett, 2003; Hu et al., 2010; Saghaei et al., 2020; Truong, 1993; Truong et al., 2008). In addition, vetiver can help reduce soil loss and runoff, which has been proven effective in mitigating soil erosion and slope stability (Babalola et al., 2007; Donjadee and Chinnarasri, 2012).

Soil column set-up

Figure 1 shows the schematic diagram of the soil column. The soil column was made of plexiglass with a total height of 1600 mm, an inner diameter of 280 mm and a wall thickness of 10 mm. The

column has two sections of the same height of 800 mm, which were connected by a pair of flange with a silicone gasket in between for preventing water leakage. The compacted soil was divided into four layers of soil from the bottom to the top, namely 400 mm clay, 200 mm gravelly sand, 400 mm slit and 300 mm CDG layers. The measured SWCCs using pressure plate apparatus (Ng and Menzies, 2007) for the four types of soils along drying and wetting paths are shown in Figure 2(a). SWCCs along the wetting path, which are frequently used to predict the hydraulic response of landfill cover under wetting condition, are fitted using Van Genuchten's (1980) model. Accordingly, based on the measured saturated hydraulic conductivities as shown in Table 1, the unsaturated hydraulic conductivities for all soils are predicted using the van Genuchten–Mualem equation (Mualem, 1976). The bottom three layers formed the new three-layer landfill cover proposed by Ng et al. (2016a), while the topsoil served as the re-cultivation layer (GEO, 2011). Before soil compaction, a layer of vacuum grease was pasted along the inner surface of the soil column to minimise any occurrence of potentially preferential flow. Then, the soils were compacted layer by layer to their targeted degree of compaction or relative density. The degree of compaction for CDG, silt and clay were 90%, 90% and 95%, respectively. The relative density of gravelly sand was 95%. The thickness of each compaction layer was 20 mm. Hence, each soil column was compacted in 65 layers in total. A trial compaction test was carried out to determine the number of blows per layer and to ensure the uniformity of the compacted soil. The variation of the measured and targeted soil density was less than 3%. Before the compaction of the successive layer, the top surface of each layer was scarified for better contact.

After completion of soil compaction, the surface of each column was sealed with plastic film for 2 days for moisture equalisation. Before transplantation, the shoot length, root depth and root characteristics (e.g. root area index, *RAI*, defined as *surface area of root per unit soil surface area*) were measured (Garg, 2015). Table 2 shows the measured characteristics of shoot length and root

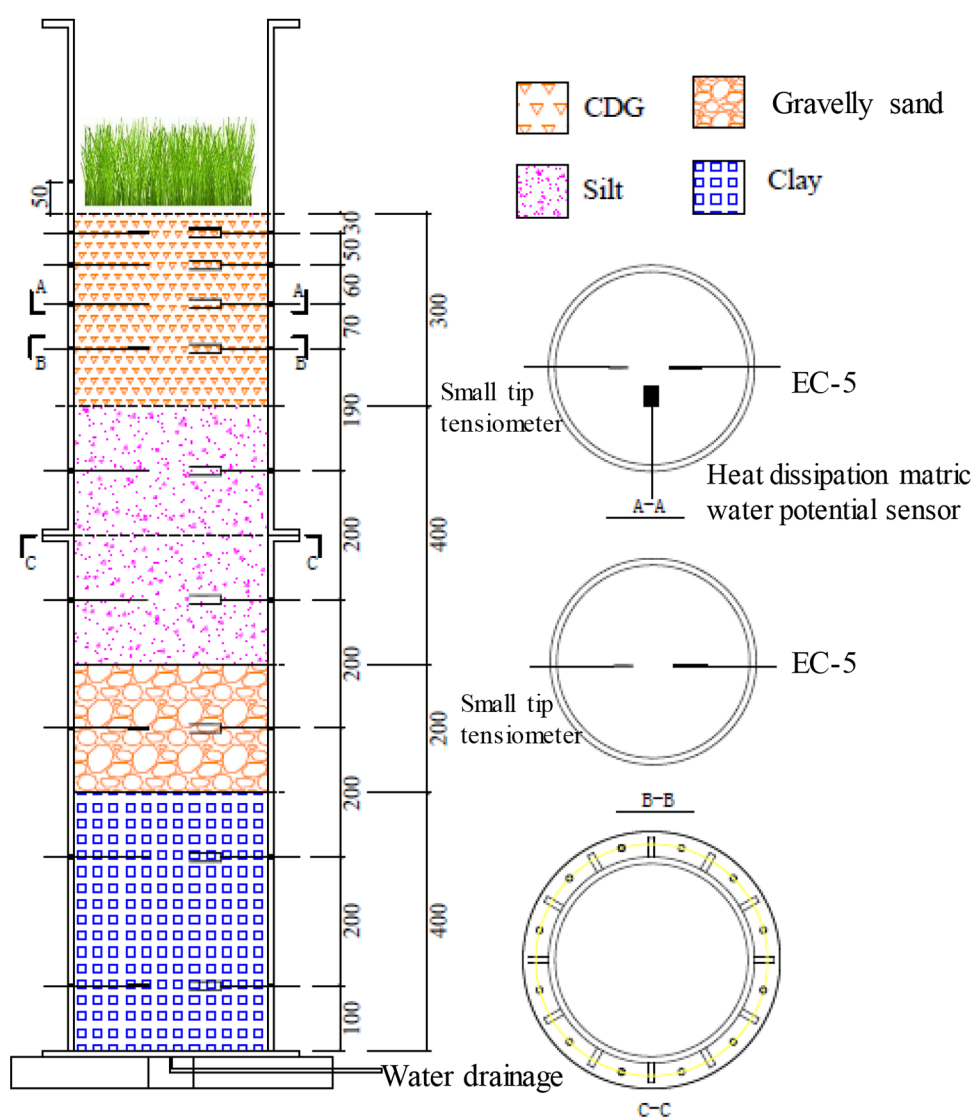


Figure 1. Schematic diagram of soil column (unit: mm).

Table 2. Summary of grass characteristics.

Grass characteristic	BG1	BG2	V1	V2
Before transplantation				
Shoot length: mm	39–50 Mean: 40 SD: ± 3	38–52 Mean: 41 SD: ± 3	280–305 Mean: 289 SD: ± 18	285–310 Mean: 295 SD: ± 15
Root depth: mm	26–33 Mean: 30 SD: ± 2	25–34 Mean: 30 SD: ± 3	85–115 Mean: 102 SD: ± 12	85–110 Mean: 100 SD: ± 10
After 40 days of growth (Adaptation period)				
Shoot length: mm	68–90 Mean: 80 SD: ± 4	70–95 Mean: 79 SD: ± 5	580–625 Mean: 600 SD: ± 35	580–610 Mean: 591 SD: ± 33
After test				
Shoot length: mm	70–90 Mean: 80 SD: ± 4	72–95 Mean: 80 SD: ± 4	650–710 Mean: 685 SD: ± 35	680–730 Mean: 700 SD: ± 33
After test				
Root depth: mm	45–60 Mean: 53 SD: ± 3	45–55 Mean: 50 SD: ± 3	170–230 Mean: 205 SD: ± 15	165–245 Mean: 200 SD: ± 18

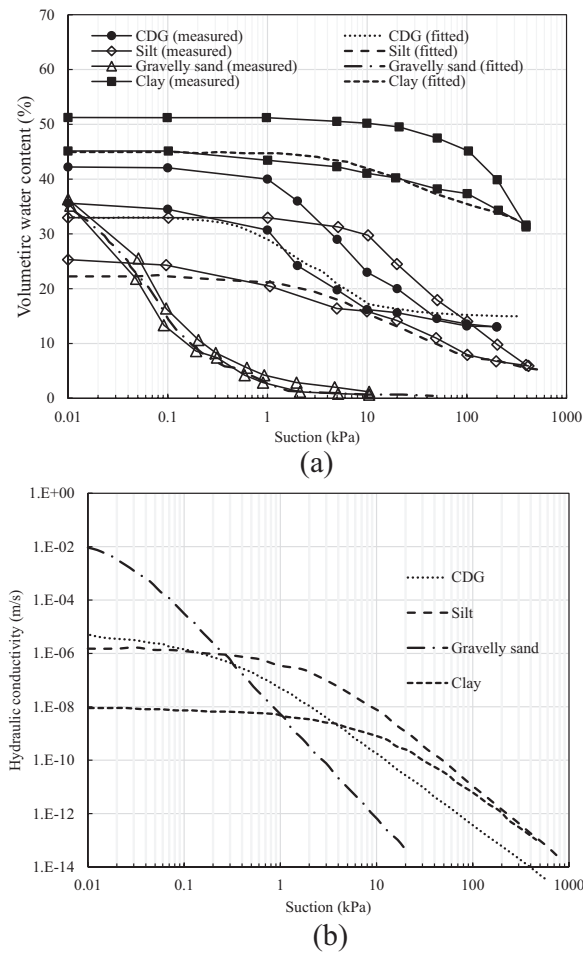


Figure 2. (a) Measured and fitted SWCCs and (b) fitted wetting hydraulic conductivity functions for CDG, silt, gravelly sand and clay.

depth. Then, Bermuda grass turf was transplanted in two soil columns. After transplantation, Bermuda turf was gently pressed to ensure close contact between the roots and the soil. Each vetiver was transplanted in another two soil columns by digging a shallow hole (e.g. 100 mm in depth and 200 mm in diameter), which was slightly larger than the root bundle, followed by refilling and compaction of the soil to the hole. According to the reported planting density of vetiver ranging from 36 plants m^{-2} to 100 plants m^{-2} (Wang, 2008), five vetivers were planted in the column (i.e. 80 plants m^{-2}). One vetiver was planted in the centre, and the other four were evenly distributed along the circumference of a circle with a diameter of 200 mm and a centre locating at the centre of the column. Then adaption period of about 40 days was conducted for plant growth, during which all the soil columns were irrigated equally about every 2–3 days. No fertiliser was added during the test to avoid any influence of different solute concentrations and hence osmotic suction (Krahn and Fredlund, 1972). During the adaptation period, any change in shoot length was monitored.

Instrumentation

Nine measurement sections were allocated along the soil column (Figure 1). In order to verify the measurements for assisting data

interpretation, EC-5 soil moisture sensors (SMs; Decagon Device Company, WA, USA) and small tip tensiometers (TMs; Soilmoisture Equipment Corporation, CA, USA) were installed at the same elevation. In the case of any possibility of cavitation of TMs during the long drying period, heat dissipation matric water potential sensors (HDs; Campbell Scientific Inc., Logan, UT, USA) were installed at the top three locations at elevations of 1160, 1220 and 1270 mm. TMs were used for measuring the low range of soil suctions (0–90 kPa), whereas HDs were used for measuring suction above 100 kPa. Before installation, all sensors were calibrated.

Test procedure

All tests were carried out in the laboratory with a controlled temperature of 23°C–25°C (i.e. average temperature in autumn in Hong Kong) and relative humidity of 46–52%. After 40 days of adaptation period, all the columns were saturated from the bottom to the top step by step, until the readings of all the tensiometers fell to 0 kPa and 100 mm ponding existed on the soil surface for at least 24 hours. Then, the ponding was drained out. Thereafter, all columns were exposed to the drying condition for 21 days, which represents light drought in Hong Kong. During the drying period, the following measurements were recorded: (1) suction and volumetric water content distributions in the three-layer landfill cover and (2) any change in shoot length of the two grass types.

During the drying process, the total moisture reduction of each column was calculated by integrating the measured water content change ($\Delta\theta$) along the depth of each column. Correspondingly, the average evapotranspiration (ET) of vegetated column was equal to the total water loss per unit area divided by time, which is shown as follows

$$ET = \frac{\int_0^h \Delta, dh}{\Delta t} \quad (1)$$

where h is the height of soil column, $\Delta\theta$ represents the volumetric water content change; and Δt is the time interval. The integration in equation (1) could be approximated by the central difference method as follows

$$ET = (h_{i+1} - h_i) \frac{(\theta_{h_i, t_0} - \theta_{h_i, t_1}) + (\theta_{h_{i+1}, t_0} - \theta_{h_{i+1}, t_1})}{2(t_1 - t_0)} \quad (2)$$

where h_{i+1} and h_i represent the $(i+1)$ th and i th measuring depth ($h_{i+1} > h_i$); t_1 and t_0 represent two measurement times ($t_1 > t_0$); θ_{h_i, t_0} and θ_{h_i, t_1} represent the measured volumetric water content at depth h_i at t_0 and t_1 , respectively; and θ_{h_{i+1}, t_0} and θ_{h_{i+1}, t_1} represent the measured volumetric water content at depth h_{i+1} at t_0 and t_1 , respectively. Similarly, transpiration (T) of bare soil can also be calculated by equation (2).

Wetting test was conducted right after the drying test to explore (1) the effects of initial suction induced by grasses on the subsequent hydrological response in the three-layer landfill cover and (2) the effectiveness of the vegetated three-layer landfill cover system on the prevention of rainfall infiltration. Before rainfall, the

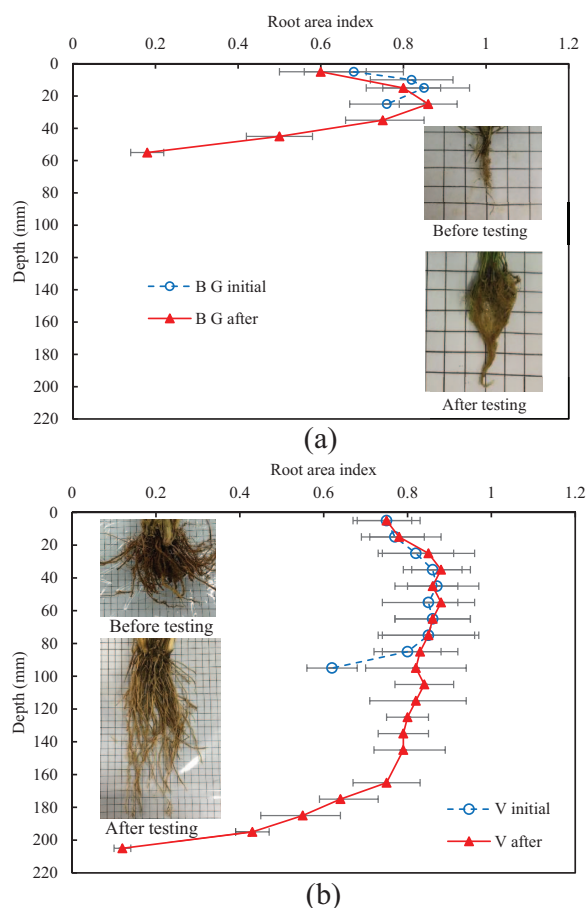


Figure 3. Measured root area index (RAI) of (a) Bermuda grass (BG) and (b) vetiver (V) before and after testing (the sheet of paper in the background of the two insets contains 10 mm × 10 mm squares).

above-ground parts of vegetation were all trimmed to avoid the influence of the above-ground parts of vegetation on rainfall ponding. During the wetting test, the drainage valves at the bottom of soil columns were open to measure water percolation. Artificial rainfall intensity of 28 mm hour⁻¹ was applied for 18 hours to simulate rainfall with 100-year return period in Hong Kong (Lam and Leung, 1995). During the wetting process, temporal variation of the ponding depth was recorded every 10 minutes for all the soil columns. As a result, the infiltration rate (I) during rainfall could be determined as follows

$$I(t) = R - \frac{1}{A} \frac{dV}{dt} \quad (3)$$

where R is the applied rainfall intensity; dt represents time interval; dV is the volume change of water ponding in the time interval of dt and A is the ponded surface area (i.e. 0.06 m²).

Suction and volumetric water content were measured every 10 minutes during rainfall. After completing the experiment, grasses were carefully dug out from the columns and soil attached to the root was washed away with care, following the standardised root washing procedure adopted by Smucker et al. (1982). Thereafter, the root length was measured, while the root characteristic (i.e. RAI) was determined by image analysis (Garg, 2015).

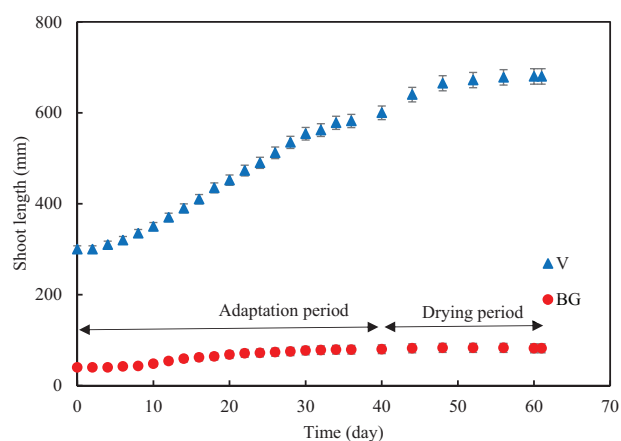


Figure 4. The development of average shoot length of Bermuda grass (BG) and vetiver (V) during adaptation and drying periods.

Measurements of the soil column tests

Development of root and shoot characteristics

Figure 3 shows roots of Bermuda grass and vetiver before and after testing. Regardless of grass type, the roots grew deeper after the test. Vetiver had longer roots than Bermuda grass. As shown by the measured RAI distributions, the root shape of both grass types did not change. The measured RAI of Bermuda grass was parabolic along the depth, while that of vetiver was almost uniform. This was because most roots of vetiver grew vertically rather than horizontally, as shown in the two insets in Figure 3(b).

Figure 4 depicts a comparison of the average shoot length of Bermuda grass and vetiver during the adaptation and drying period. After this period of growth, the vegetation coverage of Bermuda grass-planted and vetiver-planted columns is about 100% and 95%, respectively. Generally, the shoot length increased at a reduced rate. The average shoot length of Bermuda grass grew from 40 mm to about 80 mm during the adaptation period. Afterwards, its shoot length kept almost unchanged during the drying period, probably due to lack of sufficient water for plant growth. A similar trend was observed for vetiver. Yet, the shoot length of vetiver was much longer than that of Bermuda grass. Table 2 summarises the measurements of plant characteristics before and after the test.

Influence of grass type on the hydraulic response of soil during drying period

Figure 5(a) shows the measured suction profiles in the bare and vegetated soil columns during the drying period. The initial suction profiles for all the soil columns were nearly the same, which was necessary for a fair comparison. After drying for 21 days, suction for all the soil columns increased significantly. Overall, suction decreased with depth. Regardless of the grass type, the suction of

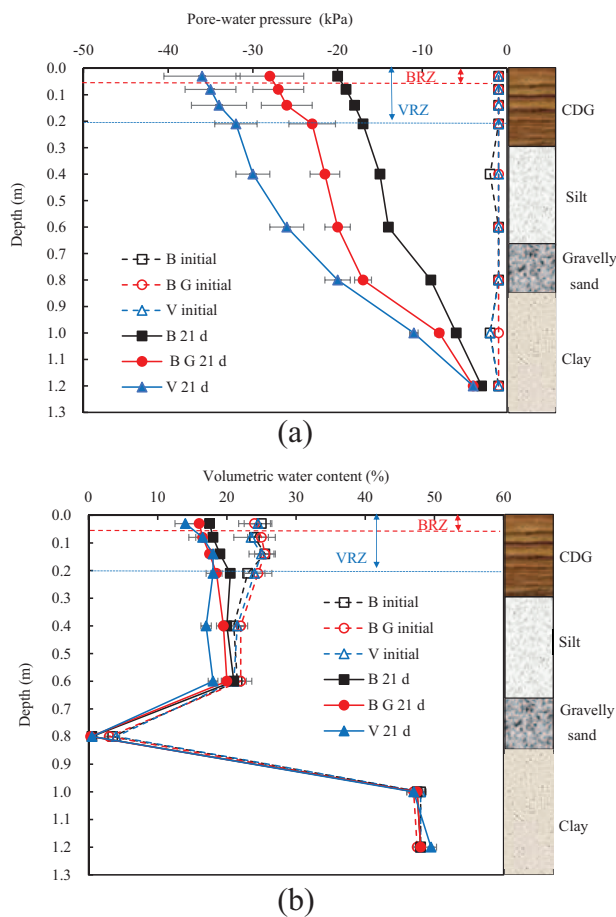


Figure 5. Hydraulic response in bare, Bermuda grass (BG)-planted and vetiver (V)-planted columns after 21 days of drying: (a) pore water pressure and (b) volumetric water content (BRZ and VRZ represent root zone of Bermuda grass and vetiver, respectively).

vegetated column was larger than that of bare soil column in the whole range of column depth. These results are consistent with those of Ng et al. (2013) and Garg et al. (2015), who found that vegetated soils had larger suction than bare soil during drying. Moreover, the suction of vetiver-planted column was larger than that of the Bermuda grass-planted case. This indicates that the vetiver-planted columns ‘lost’ more water than the Bermuda grass-planted one, probably due to vetiver having deeper roots and more shoots than those of Bermuda grass (see Figure 3) and hence more root water uptake.

Figure 5(b) shows the measured volumetric water content profiles during the drying process. Generally, measurements of volumetric water content were consistent with the suction profiles (see Figure 5(a)). After drying for 21 days, the volumetric water contents above the depth of 0.8 m in the bare soil column were larger than those in the vegetated columns. This was due to plants needing water through root water uptake to avoid dehydration (Murphy and Lodge, 2001). The overall volumetric water contents in vetiver-planted soil column were generally lower than that in the Bermuda grass planted case. This suggests that vetiver consumed more water than Bermuda grass. Moreover, negligible volumetric water content change was observed in the clay layer for all the

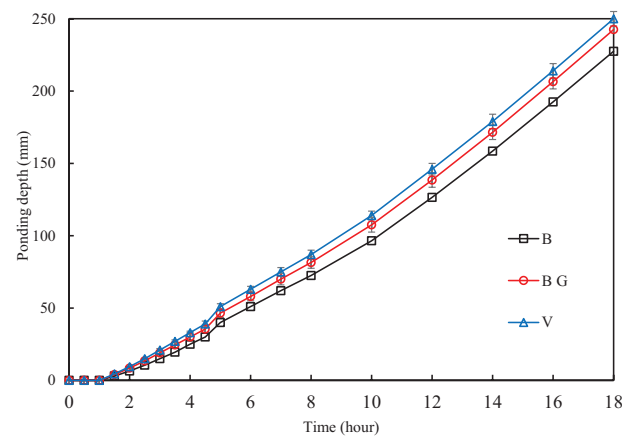


Figure 6. Measured temporal variations of ponding depth in bare, Bermuda-planted and vetiver-planted columns during artificial rainfall of 28 mm hour^{-1} for 18 hours (100-year return period).

columns. This was because the volumetric water content change was insensitive to suction, when the suction was smaller than the air entry value (i.e. 70 kPa for clay; see Figure 2(b)). These results suggest that grass with a deeper root and a larger leaf area could induce larger suction, leading to a decrease in the soil permeability and more rapid recover of water storage capacity of soils. In addition, according to equation (2), the calculated average evapotranspiration rate of vetiver planted column is about 2.4 mm day^{-1} , followed by that of Bermuda grass-planted case of 1.8 mm day^{-1} and the average evaporation rate of bare column of 1.2 mm day^{-1} . Generally, the planted column shows larger hydraulic uptake than bare soil. This is consistent with the published in-situ measurements (Garg et al., 2015; Unkovich et al., 2018), due to root water uptake. Yet, some other field tests showed that the evapotranspiration rate of bare soil and vetiver-planted case are similar (Saghaei et al., 2020). Laboratory (Cull et al., 2000; Truong et al., 2008) and long-term field experiments (Saghaei et al., 2020) were carried out to evaluate the hydraulic uptake of vetiver. Truong et al. (2008) estimated that the hydraulic uptake of vetiver ranged from 0.6 to $2.6 \text{ L plant}^{-1} \text{ day}^{-1}$ through laboratory-scale tests. During the first year of the field tests conducted by Saghaei et al. (2020), a negligible difference was observed in the measured evapotranspiration rate and water percolation between the two test sections with and without vetiver. This was likely because the vetiver was not fully developed, as the height of vetiver was only half its mature height.

Moreover, the calculated average evapotranspiration/evaporation rate is smaller than the published in-situ measurements, which range from 2 to 10 mm day^{-1} (Leung and Ng, 2013; Saghaei et al., 2020). This is probably due to less wind in the laboratory than that in the field, leading to a less evapotranspiration/evaporation rate. The calculated evapotranspiration rate of vetiver planted soil is about two times that of bare soil. This illustrates that the plant significantly increases the hydraulic water uptake and hence reduces the soil moisture contents. Therefore, less water percolation through landfill cover would be expected.

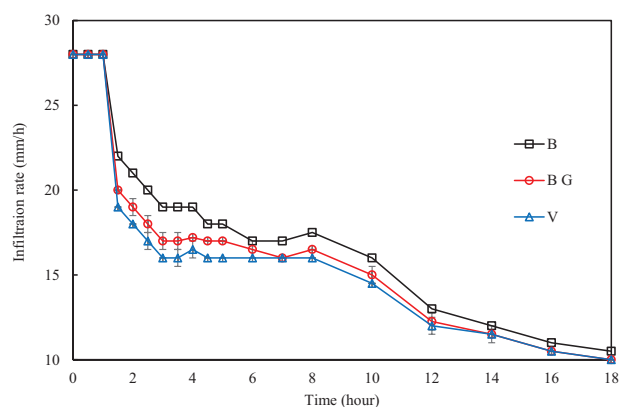


Figure 7. Calculated infiltration rates of bare, Bermuda-planted and vetiver-planted columns during artificial rainfall of 28 mm hour^{-1} for 18 hours (100-year return period).

Effects of grass type on water infiltration of soil under wetting condition

Figure 6 shows the measured temporal variation of ponding depth during the wetting process. After 1 hour of rainfall, no ponding was observed in all the columns. Afterwards, the ponding depth increased almost linearly with the rainfall duration. The ponding depth of the bare column was the lowest, suggesting the largest water infiltration. This is consistent with the experimental results of other researchers (Leung et al., 2015; Ng et al., 2016a), who found that the infiltration rate of bare soil was larger than that of vegetated soil. This was most likely due to the lower water permeability of vegetated soil caused by (1) the higher initial suction induced by root water uptake and (2) the root occupying parts of soil pores. The difference in the ponding depth of vegetated columns became more noticeable as rainfall continued. The vetiver-planted column had slightly more ponding than the Bermuda grass-planted case. This was likely attributed to that vetiver had more roots than Bermuda grass (Figure 3), resulting in more soil pore being occupied by roots, which could probably block the rainfall infiltration channels.

Figure 7 shows the temporal variation in rainfall infiltration, which was calculated by equation (1) according to the mass balance of water. During the first hour of rainfall, the infiltration rates of all the columns were almost the same, equaling to the applied rainfall intensity. Then, the infiltration rate decreased rapidly as rainfall continued. This was consistent with the measured water infiltration in bare soil given by other researchers (Morin and Benyamini, 1977; Ng et al., 2016a; Zhan et al., 2007), who found that the infiltration rate decreased exponentially with the rainfall duration. This was due to the reduction of hydraulic gradient near the ground surface, owing to the pore water pressure in deeper depths increasing as a result of the downward water flow. In contrast to the reported temporary variation of water infiltration in a homogeneous soil (Morin and Benyamini, 1977; Ng et al., 2016a; Zhan et al., 2007), the infiltration rate in the three-layer landfill cover further decreased after 8 hours of rainfall, due to the wetting front moved from the vegetated layer to the silt layer, which had a

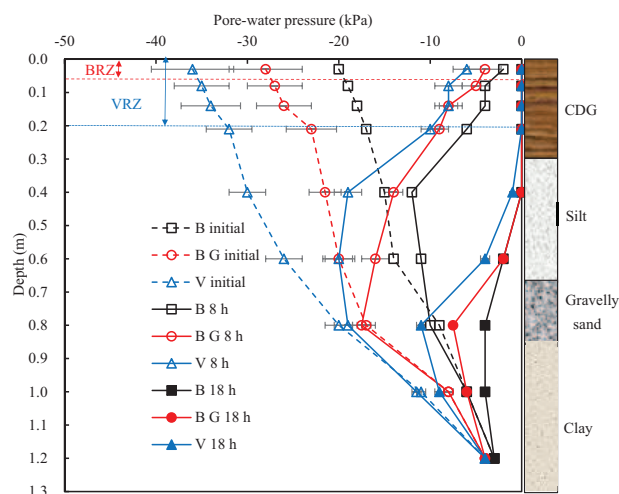


Figure 8. Measured pore water pressure distributions in bare, Bermuda-planted and vetiver-planted columns after rainfall.

lower water permeability (see Figure 8 given later). For the vegetated soils, the infiltration rates were smaller than that of the bare soil. This was because the root occupied the soil pores, thus reducing the water permeability (Leung et al., 2015; Scanlan and Hinz, 2010). For the vetiver-planted soil, the infiltration rates were always slightly lower than that in Bermuda grass covered soil. One possible reason was that vetiver has relatively more roots, which grew deeper (Figure 3), causing more soil pores to be occupied by roots and thus reducing water infiltration.

These results illustrate that the relationship between water infiltration versus rainfall duration in multi-layer soil shows a bimodal form, which is different from the unimodal one observed in a homogeneous soil (i.e. a single-layer soil). This is because of the contrasting hydraulic properties of different soil layers. The vegetated soil had relatively lower water infiltration than the bare case. Vetiver is more effective than Bermuda grass in terms of reducing water infiltration, probably due to more roots and higher initial suction induced by larger root water uptake. Yet, it should be recalled that only a relatively short-term test was conducted to simulate the grass growth stage without any observation of decay roots. Special attention should be paid to the application of the outcome to predict long-term performance (i.e. in the timescale of year) of vegetated soil, in which root decay may increase water infiltration.

Influence of grass type on hydraulic response of soil under wetting condition

Figure 8 shows the suction response during rainfall. After 8 hours of rainfall at an intensity of 28 mm hour^{-1} (i.e. 10-year return period in Hong Kong, according to Lam and Leung, 1995), the suction near the soil surface dropped significantly. The vetiver-planted column retained the largest suction among all the columns due to the largest initial suction. The suction at the soil surface in the vetiver planted column was about 30% and 50%

higher than that in Bermuda grass-planted and bare soil columns, respectively. The difference in suction distribution between the Bermuda grass-planted and vetiver-planted columns decreased as rainfall continued. After 8 hours of rainfall, the suction hardly changed at a depth of 0.8 m in the gravelly sand layer for all the columns, suggesting that the wetting front had not reached the base of the gravelly sand layer. Hence, the upper two layers (silt and gravelly sand layers) of the new three-layer landfill cover system, which formed a CCBE (Ng et al., 2016a), was effective in preventing rainfall percolation after rainfall with a 10-year return period.

After 18 hours of rainfall (i.e. 100-year return period, according to Lam and Leung, 1995), the suction decreased to zero in the CDG layer for all the columns. The differences among the vegetated and bare soil columns further diminished. However, the suction of vetiver-planted column was generally largest outside the root zone, due to the largest initial suction induced by root water uptake during the drying period. For all the columns the suction in the gravelly sand layer was significantly reduced, suggesting water breakthrough of the overlying soil layers. Moreover, vegetated soil still retained larger suction than bare soil. A similar trend was observed by Ng et al. (2013), who found that outside the root zone the suction of planted soils was higher than that of bare soil under water infiltration. During rainfall, no percolation was observed for all the columns, which demonstrated that the new three-layer landfill cover system performed satisfactorily under extreme rainfall (Ng et al., 2016a). These results suggest that vetiver has a stronger capacity than Bermuda grass to enhance the effectiveness of the three-layer landfill cover in terms of reducing water breakthrough, due to (1) the stronger root water uptake ability of vetiver and (2) more roots of vetiver and hence a reduced soil water permeability caused by the root occupancy of soil pores.

Figure 9 shows a comparison of the measured volumetric water contents between bare and vegetated soil columns during rainfall. Generally, the response of volumetric content was consistent with that of suction (Figure 8). Before rainfall, the volumetric water contents of the vegetated soil columns were lower than that of the bare column due to root water uptake. During rainfall, more noticeable increase in volumetric water content could be observed for the top three layers. By contrast, negligible volumetric water content change was observed below a depth of 1 m in the clay layer, even after 18 hours of rainfall (i.e. corresponding to rainfall with 100-year return period in Hong Kong), illustrating that the three-layer landfill cover was effective in preventing water breakthrough even under rainfall with a 100-year return period. Regardless of the grass types, it is interesting that the measured volumetric water content in the CDG layer of the vegetated columns was slightly larger than that of bare soil. This could be due to the reduction in pore size due to root occupation, which increased the water retaining capacity of the soil (Ng et al., 2016c). After 18 hours of rainfall, the volumetric water content of the gravelly sand layer in the bare column was the largest among all columns, indicating the

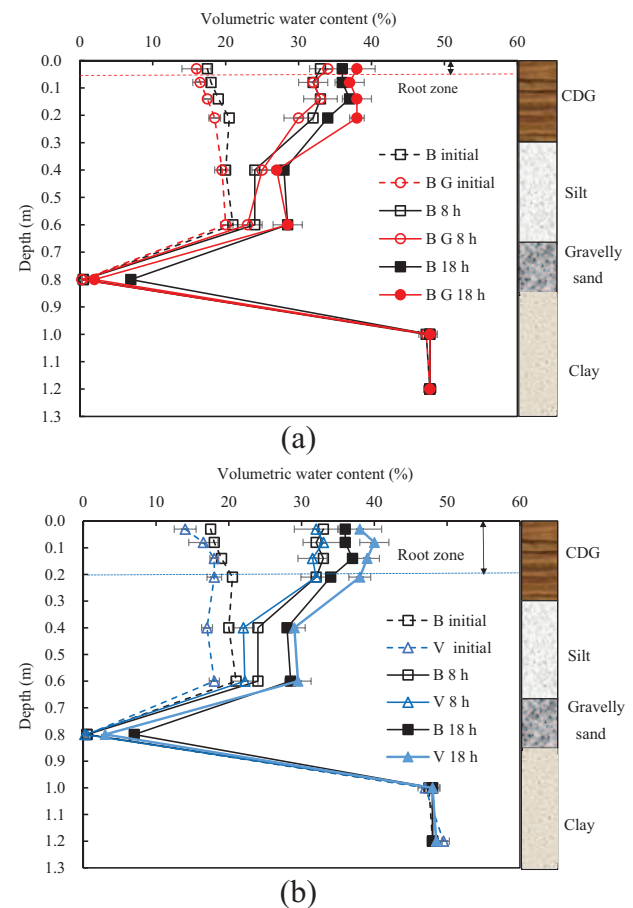


Figure 9. Comparisons of measured volumetric water contents between: (a) bare and Bermuda-planted columns. (b) bare and vetiver-planted columns after rainfall.

deepest wetting front. Nevertheless, no water percolation was measured at the base of all the columns. These findings illustrate that grass could enhance the effectiveness of the new three-layer landfill cover system under extreme rainfall condition by reducing the soil moisture during the drying period before rainfall. It should be noted that the present study was carried out in the laboratory with control temperature and humidity and only considered a relatively short-term growth period (about 2 months). That is, the test conditions could be significantly different from the field. Thus, the vetiver might not fully grow as that would exist in long-term field tests. Yet, the evapotranspiration rate of vetiver was expected to be affected by the vetiver characteristics (e.g. root-ball mass). Accordingly, caution should be paid to extrapolate the results of the present laboratory-scale tests to the field-scale tests. In addition, it should also be remembered that only the short-term growth period (about 2 months) of grass was considered. Accordingly, the root water uptake increases as root that grows in long-term period cannot be captured. Therefore, special attention should be paid when extrapolating the conclusions of the present study to the long-term prediction of hydraulic performance of vegetated landfill cover, as root may decay and hence generate preferential flow during the long-term period.

Summary and conclusion

Soil column tests were conducted to investigate the effects of grass type on the hydraulic response of the three-layer landfill cover system. Bermuda grass and vetiver were selected.

During drying, the vetiver-planted soil column had the largest suction, followed by the Bermuda grass-planted case and bare soil column. Specifically, the suction near the soil surface of vetiver plant columns was about 8 kPa larger than that of Bermuda grass-planted cases. This was attributed to the larger *RAI* and larger leaf area of vetiver than those of Bermuda grass, leading to more root water uptake.

During the wetting period, the largest infiltration rate existed in the bare soil column, followed by the Bermuda grass-planted and vetiver-planted cases. This was probably due to the relatively larger suction before rainfall and hence a lower water permeability. Under rainfall with 10-year return period, the vetiver-planted soil retained relatively larger suction than that of Bermuda grass-planted case in the silt layer. These results suggest that vetiver has a stronger capacity than Bermuda grass to enhance the effectiveness of the three-layer landfill cover in terms of reducing water breakthrough, due to its larger root amounts and deeper root, thus (1) inducing larger initial suction through root water uptake during the drying period and (2) reducing soil water permeability through the occupation of soil pores by roots. Even after rainfall with a 100-year return period, no water percolation was observed for all the soil columns, suggesting that the three-layer landfill cover is effective in preventing water percolation in a humid climate.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship and/or publication of this article: This project was supported by the National Key Research and Development Program of China (2019YFC1806003), and National Natural Science Foundation of China (Grant Nos 51908134, 51808125, 52178320, 42177120, 41861134011, U2005205 and 51778249). S.F. thanks the Foundation of Key Laboratory of Soft Soils and Geoenvironmental Engineering (Zhejiang University), Ministry of Education, for providing funding (Grant No. 2019P06). H.W. Liu thanks the research funding provided by Key Laboratory of Geohazard Prevention of Hilly Mountains, Ministry of Natural Resources (Fujian Key Laboratory Of Geohazard Prevention) (Grant No. FJKLGH2021K003).

ORCID iDs

Song Feng  <https://orcid.org/0000-0002-0742-9980>

Hong Wei Liu  <https://orcid.org/0000-0001-7116-9952>

References

ASTM (2006) Standard test method for permeability of granular soils (constant head). ASTM D2434, USA.
ASTM (2010a) Standard test method for specific gravity of soil solids by water pycnometer. ASTM D854, USA.

ASTM (2010b) Standard test methods for liquid limit, plastic limit, and plasticity index of soils. ASTM D4318, USA.
ASTM (2010c) Standard test methods for measurement of hydraulic conductivity of saturated porous materials using a flexible wall permeameter. ASTM D5084, USA.
ASTM (2011) Standard practice for classification of soils for engineering purposes (Unified Soil Classification System). ASTM D2487, USA.
ASTM (2012) Standard test method for laboratory compaction characteristics of soil using standard effort. ASTM D698, USA.
Babalola O, Oshunsanya SO and Are K (2007) Effects of vetiver grass (*Vetiveria nigriflora*) strips, vetiver grass mulch and an organomineral fertilizer on soil, water and nutrient losses and maize (*Zea mays*, L) yields. *Soil and Tillage Research* 96: 6–18.
Beyrouthy CA, West C and Gbur EE (1990) Root development of Bermuda grass and tall fescue as affected by cutting interval and growth regulators. *Plant and Soil* 127: 23–30.
Białowiec A, Wojnowska-Baryła I and Hasso-Agopsowicz M (2007) The controlling of landfill leachate evapotranspiration from soil-plant systems with willow: *Salix amygdalina* L. *Waste Management & Research* 25: 61–67.
Camerini G and Groppali R (2014) Landfill restoration and biodiversity: A case of study in northern Italy. *Waste Management & Research* 32: 782–790.
Cull RH, Hunter H, Hunter M, et al. (2000) Application of Vetiver grass technology in off-site pollution control. II. Tolerance to herbicides under selected wetland conditions. In: *2nd international conference on Vetiver*, Phetchaburi, Thailand, 18–22 January.
Donjatee S and Chinnarasri C (2012) Vetiver grass mulch for prevention of runoff and soil loss. *Agricultural Water Management* 166: 144–151.
Fatahi B, Khabbaz H and Indraratna B (2009) Parametric studies on bioengineering effects of tree root-based suction on ground behavior. *Ecological Engineering* 35: 1415–1426.
Feng S, Liu HW and Ng CWW (2019) Analytical solutions for one-dimensional water flow in vegetated layered soil. *International Journal of Geomechanics* 19: 04018191.
Feng S, Liu HW and Ng CWW (2020) Analytical analysis of the mechanical and hydrological effects of vegetation on shallow slope stability. *Computers and Geotechnics* 118: 103335.
Gadi VK, Bordoloi S, Garg A, et al. (2018) Effect of shoot parameters on cracking in vegetated soil. *Environmental Geotechnics* 5: 123–130.
Garg A (2015) *Effects of vegetation types and characteristics on induced soil suction*. PhD Thesis, The Hong Kong University of Science and Technology, Pok Fu Lam, Hong Kong.
Garg A, Leung AK and Ng CWW (2015) Comparisons of soil suction induced by evapotranspiration and transpiration of *S. heptaphylla*. *Canadian Geotechnical Journal* 52: 2149–2155.
GEO (2011) Technical guidelines on landscape treatment for slopes. Hong Kong. Available at: <https://www.cedd.gov.hk/eng/publications/geo/geo-gco/geo-p111/index.html>
Hau BC and Corlett RT (2003) Factors affecting the early survival and growth of native tree seedlings planted on a degraded hillside grassland in Hong Kong, China. *Restoration Ecology* 4: 483–488.
Hu L, Wang Z, Du H, et al. (2010) Differential accumulation of dehydrins in response to water stress for hybrid and common Bermudagrass genotypes differing in drought tolerance. *Journal of Plant Physiology* 167: 103–109.
Khalilzad M, Gabr MA and Hynes ME (2014) Effects of woody vegetation on seepage-induced deformation and related limit state analysis of levees. *International Journal of Geomechanics* 14: 302–312.
Krahn J and Fredlund DG (1972) On total, matric and osmotic suction. *Soil Science* 114: 339–348.
Lam CC and Leung YK (1995) *Extreme rainfall statistics and design rain-storm profiles at selected locations in Hong Kong*. Technical Note No. 86. Kowloon, Hong Kong: Royal Observatory.
Leung AK and Ng CWW (2013) Analyses of groundwater flow and plant evapotranspiration in a vegetated soil slope. *Canadian Geotechnical Journal* 50: 1204–1218.
Leung AK, Garg A, Coo JL, et al. (2015) Effects of the roots of *Cynodon dactylon* and *Schefflera heptaphylla* on water infiltration rate and soil hydraulic conductivity. *Hydrological Processes* 29: 3342–3354.

- Liang Z, Liu H, Zhao Y, et al. (2020) Effects of rainfall intensity, slope angle, and vegetation coverage on the erosion characteristics of Pisha sandstone slopes under simulated rainfall conditions. *Environmental Science and Pollution Research* 27: 17458–17467.
- Liu HW, Feng S and Ng CWW (2016) Analytical analysis of hydraulic effect of vegetation on shallow slope stability with different root architectures. *Computers and Geotechnics* 80: 115–120.
- Liu HW, Feng S, Garg A, et al. (2018) Analytical solutions of pore-water pressure distributions in a vegetated multi-layered slope considering the effects of roots on water permeability. *Computers and Geotechnics* 102: 252–261.
- Morin J and Benyamini Y (1977) Rainfall infiltration into bare soils. *Water Resources Research* 13: 813–817.
- Mualem Y (1976) A new model for predicting the hydraulic conductivity of unsaturated porous media. *Water Resources Research* 12: 513–522.
- Murphy SR and Lodge GM (2001) Plant density, litter and bare soil effects on actual evaporation and transpiration in autumn. In: *Proceedings of the 10th Australian agronomy conference*, Hobart, TAS, Australia, 29 January.
- Ng CWW and Leung AK (2012) Measurements of drying and wetting permeability functions using a new stress-controllable soil column. *Journal of Geotechnical and Geoenvironmental Engineering* 138: 58–68.
- Ng CWW and Menzies B (2007) *Advanced Unsaturated Soil Mechanics and Engineering*. New York: Taylor & Francis.
- Ng CWW and Zhan LT (2007) Comparative study of rainfall infiltration into a bare and a grassed unsaturated expansive soil slope. *Soils and Foundations* 47: 207–217.
- Ng CWW, Chen R, Coo JL, et al. (2019a) A novel vegetated three-layer landfill cover system using recycled construction wastes without geomembrane. *Canadian Geotechnical Journal* 56: 1863–1875.
- Ng CWW, Coo JL, Chen ZK, et al. (2016a) Water infiltration into a new three-layer landfill cover system. *Journal of Environmental and Engineering Geophysics* 142: 04016007.
- Ng CWW, Garg A, Leung AK, et al. (2016b) Relationships between leaf and root area indices and soil suction induced during drying–wetting cycles. *Ecological Engineering* 91: 113–118.
- Ng CWW, Liu HW and Feng S (2015) Analytical solutions for calculating pore water pressure in an infinite unsaturated slope with different root architectures. *Canadian Geotechnical Journal* 52: 1981–1992.
- Ng CWW, Lu BW, Ni JJ, et al. (2019b) Effects of vegetation type on water infiltration in a three-layer cover system using recycled concrete. *Journal of Zhejiang University-SCIENCE A* 20: 1–9.
- Ng CWW, Ni JJ, Leung AK, et al. (2016c) Effects of planting density on tree growth and induced soil suction. *Géotechnique* 66: 711–724.
- Ng CWW, Woon KX, Leung AK, et al. (2013) Experimental investigation of induced suction distribution in a grass-covered soil. *Ecological Engineering* 52: 219–223.
- Ni JJ, Leung AK, Ng CWW, et al. (2018) Modelling hydro-mechanical reinforcements of plants to slope stability. *Computers and Geotechnics* 95: 99–109.
- Preston GM and McBride RA (2004) Assessing the use of poplar tree systems as a landfill evapotranspiration barrier with the SHAW model. *Waste Management & Research* 22: 291–305.
- Saghaei B, Khire MV, Caldwell M, et al. (2020) Field-scale evaluation of evapotranspiration from a landfill cover. In: *Geo-congress 2020: Modeling, geomaterials, and site characterization*, Reston, VA, 25 February.
- Scanlan CA and Hinz C (2010) Insights into the processes and effects of root-induced changes to soil hydraulic properties. In: *19th world congress of soil science, soil solutions for a changing world*, Brisbane, QLD, Australia, pp.41–44. Available at: <https://www.iuss.org/19th%20WCSS/Symposium/pdf/0247.pdf>
- Singh SR, Prakash A, Garg A, et al. (2018) Stochastic modeling of relative permeability for vegetated covers. *International Journal of Geomechanics* 18: 06018020.
- Smethurst JA, Briggs KM, Powrie W, et al. (2015) Mechanical and hydrological impacts of tree removal on a clay fill railway embankment. *Géotechnique* 65: 869–882.
- Smethurst JA, Clarke D and Powrie W (2006) Seasonal changes in pore water pressure in a grass-covered cut slope in London clay. *Géotechnique* 56: 523–537.
- Smucker AJM, McBurney SL and Srivastava AK (1982) Quantitative separation of roots from compacted soil profiles by the hydropneumatic elutriation system 1. *Agronomy Journal* 74: 500–503.
- Truong P (1993) Report on the international vetiver grass field workshop, Kuala Lumpur. *Australian Journal of Soil and Water Conservation* 6: 23–26.
- Truong P, Van TT and Pinnars E (2008) Vetiver system applications technical reference manual. *Vetiver Network International*. Available at: http://www.vetiver.org/TVN-Manual_Vf.pdf
- Unkovich M, Baldock J and Farquharson R (2018) Field measurements of bare soil evaporation and crop transpiration, and transpiration efficiency, for rainfed grain crops in Australia – A review. *Agricultural Water Management* 205: 72–80.
- Van Genuchten MT (1980) A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Science Society of America Journal* 44: 892–898.
- Wang WH (2008) Study on cultivation techniques of vetiver grass in highway slope of Karst mountainous region. Guizhou University, Guiyang, China (in Chinese).
- Woon KX (2013) *Field and laboratory investigations of Bermuda grass induced suction and distribution*. MPhil Thesis, The Hong Kong University of Science and Technology, Pok Fu Lam, China.
- Zhan LT, Ng CWW and Fredlund DG (2007) Field study of rainfall infiltration into a grassed unsaturated expansive soil slope. *Canadian Geotechnical Journal* 44: 392–408.

Appendix 1

Notation

A	ponded surface area of column
e	void ratio
G_s	specific gravity
h	height of soil column
I	infiltration rate
k_s	saturated water permeability
R	rainfall intensity
t	time
V	volume of ponding water
w	volumetric water content at saturation
ρ_d	maximum dry density
θ	volumetric water content